Case Study 1: Attendance Recovery Computation

Assignment Case No. 12 | Level: Beginner

01 PROBLEM STATEMENT

Calculate current attendance percentages and determine the minimum consecutive classes required to reach an eligibility threshold under different future attendance scenarios.

Problem interpretation: A student needs to know whether their attendance meets the required percentage. If it does not, calculate how many consecutive classes they must attend, assuming they attend every upcoming class.

02 OBJECTIVES

Calculate attendance percentage from classes attended and conducted.

Check whether the eligibility threshold has been met.

Find the minimum future classes needed to reach the threshold.

03 ASSUMPTIONS

Attendance is represented by whole numbers.

The student attends every future class being calculated.

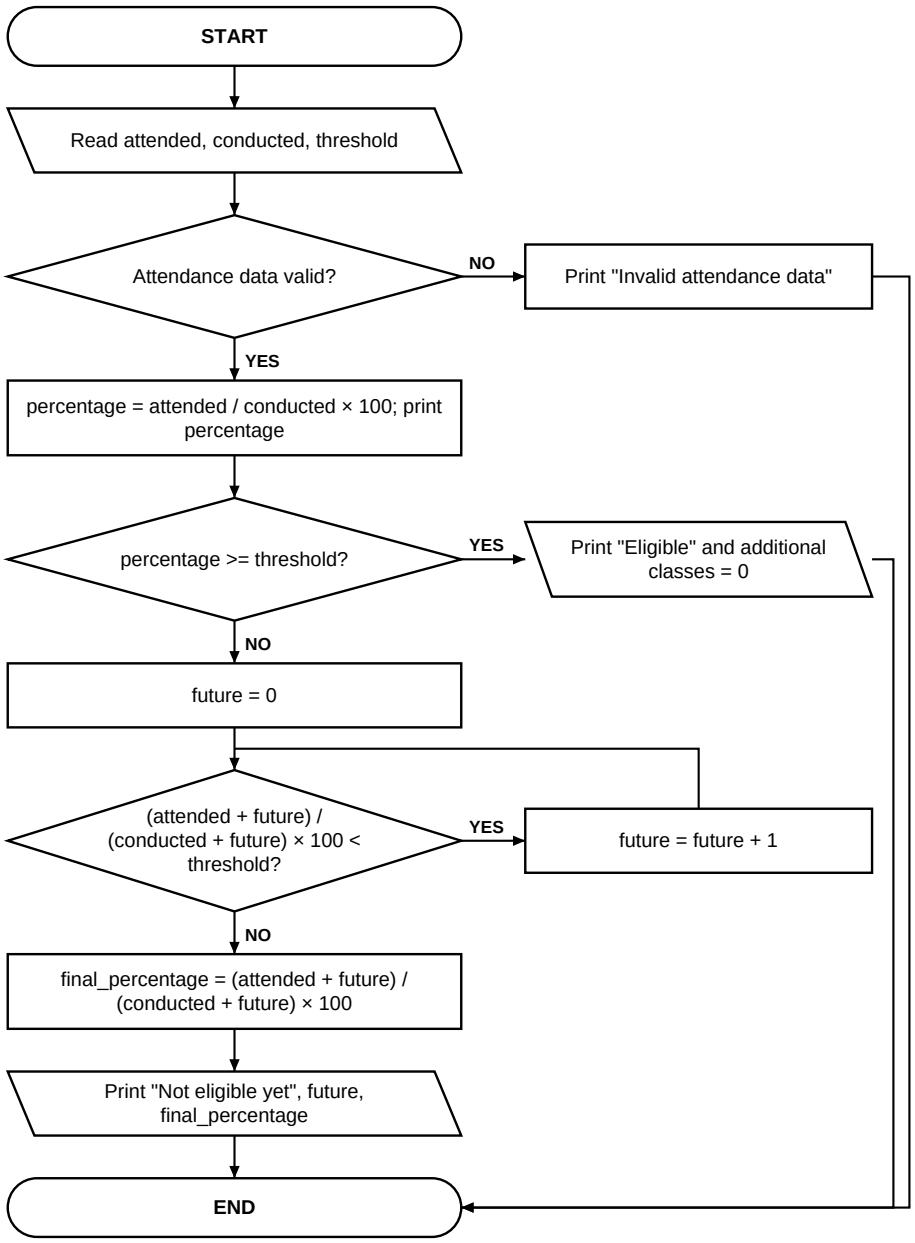
The required attendance threshold is 75%.

Classes conducted must be greater than zero and attended classes cannot exceed conducted classes.

04 ALGORITHM

1. Start.
2. Set classes attended, classes conducted, and the required percentage.
3. Validate the attendance values.
4. Calculate current percentage = attended / conducted × 100.
5. If the percentage meets the threshold, report eligibility.
6. Otherwise, increase the future-class count one by one until the percentage meets the threshold.
7. Display the percentage, eligibility, and minimum classes needed.
8. Stop.

05 FLOWCHART (TEXT FLOW)



06 PSEUDOCODE

```
BEGIN
    attended = 42; conducted = 60; threshold = 75
    IF conducted <= 0 OR attended < 0 OR attended > conducted THEN
        PRINT "Invalid attendance data"
    ELSE
        percentage = attended / conducted * 100
        PRINT percentage
        IF percentage >= threshold THEN
            PRINT "Eligible"
            PRINT additional classes needed = 0
        ELSE
            future = 0
            WHILE (attended + future) / (conducted + future) * 100 < threshold
                future = future + 1
            END WHILE
            PRINT "Not eligible yet", future
            final_percentage = (attended + future) / (conducted + future) * 100
            PRINT final_percentage
        END IF
    END IF
END
```

07 PYTHON SOURCE CODE

The complete Python program is provided in the separate file 01_Attendance_Recovery_Computation_Code.pdf.

08 TEST CASES

INPUT / SCENARIO	CHECK	EXPECTED RESULT
42 attended / 60 conducted	70.00%, below 75%	Not eligible yet; 12 classes needed; 75.00% afterwards
45 attended / 60 conducted	75.00% (boundary)	Eligible; additional classes needed: 0
61 attended / 60 conducted	Attended exceeds conducted	Invalid attendance data
0 attended / 0 conducted	Conducted is not above zero	Invalid attendance data

09 SAMPLE OUTPUT PREVIEW

```
Current attendance: 70.00%
Status: Not eligible yet
Additional consecutive classes to attend: 12
Attendance after those classes: 75.00%
```

10 CONCLUSION

The program calculates the current attendance percentage and uses a loop to find the smallest number of consecutive classes needed to reach the eligibility threshold.